

Damien Robert

Deep Learning for Remote Sensing & Environment

Summary I am a postdoctoral researcher in the EcoVision lab at University of Zurich, collaborating with Jan D. Wegner. I am broadly interested in deep learning for real-world data and impactful applications, with a taste for approaches making deep learning research socially and environmentally beneficial, accessible and reproducible. My recent work focuses on efficient and scalable deep learning, forest structure analysis, and species distribution modeling.



Positions

2024 – Now : *Postdoctoral Researcher, EcoVision lab, University of Zurich*

Deep learning for remote sensing and environment

PI: Jan D. Wegner

2020 – 2024 : *PhD student, ENGIE Lab CRIGEN - LASTIG, IGN/ENSG*

3y 4m

Efficient learning on large-scale 3D point clouds

Advisors: Loïc Landrieu and Bruno Vallet

2017 - 2020 : *R&D Engineer, SIRADEL, ENGIE*

2y 8m

Deep Learning on large-scale, terrestrial/aerial, indoor/outdoor 3D/2D data

2017 : *Co-Founder, Inspirama*

1y

Website gathering book recommendations from inspiring people

2015 : *R&D Intern, Dassault Systemes*

6m

Dimensionality reduction and dynamic system modeling

2014 : *R&D Intern, Dassault Systemes*

6m

UX design

Education

2022 : *International Computer Vision Summer School*

Sicily, Italy

CV courses by world-renowned experts in academia and industry

2011 - 2015 : *Ecole Centrale Lyon, MSc*

Lyon, France

Mathematics, Computer Science, Mechanics, Signal Processing, Automation

2017 : *CNRS AI Fall School*

Lyon, France

Multi-disciplinary course for AI students and researchers

2017 : *Udacity, Machine Learning Engineer Nanodegree*

MOOC

Machine learning, mathematics, computer science

2015 : *Coursera*

MOOC

Introduction to Machine Learning

2009 - 2011 : *Chateaubriand High School*

Rennes, France

Preparation course for exams to enter French engineering schools

2006 - 2009 : *Victor & Helene Basch High School*

Rennes, France

High School Diploma with honours, specialized in Sciences and English

Research Experience



Publications

2025

CVPR workshop EarthVision: Elena Plekhanova, [Damien Robert](#), Johannes Dollinger, Emilia Arens, Philipp Brun, Jan Dirk Wegner, Niklaus Zimmermann, *SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology*

IEEE Geoscience and Remote Sensing Magazine: Carlos Gomes, Isabelle Wittmann, [Damien Robert](#), et al., *Lossy neural compression for geospatial analytics: A review*

ICLR workshop Tackling Climate Change with Machine Learning: Johannes Dollinger, [Damien Robert](#), Elena Plekhanova, Lukas Drees, Jan Dirk Wegner, *Climplicit: Climatic Implicit Embeddings for Global Ecological Tasks*

ISPRS Geospatial Week Oral: Kaan Karaman, Yuchang Jiang, [Damien Robert](#), Vivien Sainte Fare Garnot, Maria João Santos, Jan Dirk Wegner, *GSR4B: Biomass Map Super-Resolution with Sentinel-1/2 Guidance*

2024

PhD Thesis: [Damien Robert](#), *Efficient Learning on Large-Scale 3D Point Clouds*. **Jury:** Sébastien Lefèvre, Cédric Demonceaux, Patrick Pérez, Siyu Tang, Duygu Ceylan, Loic Landrieu, Bruno Vallet

3DV Oral (top 5.3% submissions): [Damien Robert](#), Hugo Raguét, Loic Landrieu, *Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering*

2023

ICCV (top 26.8% submissions): [Damien Robert](#), Hugo Raguét, Loic Landrieu, *Efficient 3D Semantic Segmentation with Superpoint Transformer*

2022

CVPR Best paper finalist (top 0.4% submissions): [Damien Robert](#), Bruno Vallet, Loic Landrieu, *Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation*



Supervision & Collaboration

PhD Students

Emilia Arens with Jan D. Wegner (UZH)

Johannes Dollinger with Jan D. Wegner (UZH)

Kaan Karaman with Jan D. Wegner (UZH)

Louis Geist with Loic Landrieu (ENPC)

Yuanwen Yue with Konrad Schindler (ETH) and Christian Rupprecht (Oxford)

Post-Doctoral Fellows

Elena Plekhanova (WSL)

Master Students

Valerio Schelbert (ETH)

Jackson Sunny (Polytechnique) with SAMP

Consulting

UpCircle AI (Switzerland)

Ecole des Sciences Criminelles de Lausanne (Switzerland)

SAMP AI (France)

GEODIGITAL (US)



Reviewing

2025

CVPR, CVPR workshop Earth Vision

2024

ISPRS Journal of Photogrammetry and Remote Sensing, ICLR workshop ML4RS, ECCV (outstanding reviewer), CVPR workshop Earth Vision

2023

CVPR, CVPR workshop Earth Vision

2022

ISPRS Journal of Photogrammetry and Remote Sensing, CVPR workshop Earth Vision

Teaching

2025 : UZH

(M2 - 18 hours)

Course on Attention & AI4Good group projects

2024 : UZH

(M2 - 5.5 hours)

Course and labs on Attention, NeRFs, and Diffusion

2023 : ENSG-IGN

(M2 - 13 hours)

Course and labs on Deep Learning for Remote Sensing

2022 : XXIV ISPRS Congress

(Researchers - 1 day)

Tutorial on Deep Learning for Remote Sensing

2022 : ENGIE CRIGEN lab

(Researchers - 1 day)

Tutorial on 3D Deep Learning, Torch-Points3D & DeepViewAgg

2022 : ENSG-IGN

(M2 - 9 hours)

Course and labs on Deep Learning for Remote Sensing

2020 : Ecole Polytechnique

(M1 - 12 hours)

Course on Deep Learning for Computer Vision

Open-Source Repositories

 drprojects/superpoint_transformer	880 ★	118 🍴
 drprojects/DeepViewAgg	234 ★	24 🍴
 drprojects/point_geometric_features	68 ★	6 🍴
 drprojects/nora	45 ★	2 🍴

Conferences and Invited Talks

 Conference oral  Poster  Invited talk  Interview

2025

AI4EO

Rennes, France

SSL4Eco: A Global Seasonal Dataset for Geospatial Foundation Models in Ecology

COMET TSI (CNES)

Virtual

Remote Sensing for Species Distribution Modeling

2024

3D Data Academy

YouTube

Interview by Florent Poux and tutorial for Superpoint Transformer

3DV

Davos, Switzerland

Scalable 3D Panoptic Segmentation as Superpoint Graph Clustering

2023

National Land Survey of Finland (NLS)

Paris, France

Presenting IGN's research on large-Scale 2D and 3D Learning

 Ecole des Ponts, IMAGINE lab	Paris, France
Efficient Learning on Large-Scale 3D Point Clouds	
 ICCV	Paris, France
Efficient 3D Semantic Segmentation with Superpoint Transformer	
 ETH Zurich, Computer Vision and Geometry lab	Zurich, Switzerland
Efficient Learning on Large-Scale 3D Point Clouds	
 ETH Zurich, Photogrammetry and Remote Sensing lab	Zurich, Switzerland
Efficient Learning on Large-Scale 3D Point Clouds	
 ENGIE CRIGEN lab	Paris, France
Efficient 3D Semantic Segmentation with Superpoint Transformer	
 Samp R&D lab	Paris, France
Efficient 3D Semantic Segmentation with Superpoint Transformer	
 University of Zurich, EcoVision lab	Virtual
Efficient 3D Semantic Segmentation with Superpoint Transformer	
 Valeo.ai	Paris, France
Efficient 3D Semantic Segmentation with Superpoint Transformer	

2022

 IGN, LASTIG lab	Paris, France
Self-Supervised Learning for Computer Vision	
 Bundesamt für Kartographie und Geodäsie (BKG)	Paris, France
Presenting IGN's research on Large-Scale 2D and 3D Learning	
 International Computer Vision Summer School	Sicily, Italy
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	
 CV News	New Orleans, US
Interviewed by the CV News journal for its <i>Best of CVPR'22</i> issue	
 CVPR	New Orleans, US
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	
 Ecole des Ponts, IMAGINE lab	Paris, France
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	
 XXIV ISPRS Congress	Nice, France
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	
 Ecole Polytechnique, LIX lab	Paris, France
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	
 AI4GEO project seminar	Virtual
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	
 IGN-ENSG Research Days	Paris, France
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	
 GDR ISIS seminar	Virtual
Learning Multi-View Aggregation In the Wild for Large-Scale 3D Semantic Segmentation	

2021

 IGN-ENSG Research Days	Virtual
Multimodal learning on point clouds and images	

Skills, Interests, and Personal

Research Topics

- Computer vision
- Deep learning
- LiDAR data
- Large-scale 3D data
- Multimodal learning
- Efficient learning
- Superpoint-based learning
- Graph-based learning
- Graph partitioning
- Self-supervised learning
- Remote sensing
- Forest structure analysis
- Species distribution modeling
- Representation learning for macroecology
- EO data representation and compression

Tools

-  Python
-  PyTorch
-  PyTorch Lightning
-  PyTorch Geometric
-  Hydra
-  scikit-learn
-  Plotly
-  Weights & Biases
-  Blender
-  C++
-  Git
-  LaTeX
-  Linux

Languages

-  French Native
-  English Fluent
-  Spanish Intermediate
-  German Beginner

International Experience

- 2024-Now  Zurich
- 2015-2016  Backpacking
- 2014-2015  Providence, RI

Personal Interests

